

TRINITY

A Concise 100-Point Walkthrough, with an Opcode Census

Infocom, 1986 (stories/trinity.z4, Z-machine version 4), played on the zAI-Machine interpreter with random seed 42. Final result: **100 of a possible 100, in 600 moves**, ending as a *Tourist* back at Palace Gate. Every command below is replayed by `TrinityWalkthroughTests.TrinityPerfectScore`, which also tallies each opcode executed (see the Opcode Census at the end).

Solved during play with the game's manual (the sundial, the Trinity site map and the crane instructions) as the only outside reference. No published walkthrough or hints were consulted. The interpreter's disassembler and object/global inspection tooling were used to understand puzzles.

Know Before You Go

- **The sundial is the map.** From the Vertex (Bottom of Stairs, N, N), PUT GNOMON IN HOLE, then TURN RING TO *n* to pick a symbol. The triangle's shadow sweeps across the hub, and a toadstool's white door opens when the shadow's point touches it. PULL LEVER unfreezes the sun. Wait until the ring clicks, then PUSH LEVER to freeze the door open. Each door works once, and walking back out of its white door ends the visit.
- **Symbols used (ring position → door → destination):** 1 → Waterfall → Earth orbit; 2 → Ossuary → Nevada (underground test); 3 → Mesa → Eniwetok (Mike shot); 4 → Herb Garden → Siberia; 5 → Moor → Nagasaki; 6 → Islet → Trinity site. Ring 0 (the Meadow door) is where you arrived from London.
- **The gnomon is backwards.** The London gnomon won't fit until it is mirrored by walking the Arboretum pergola (N, U, S, D from Top of Arbor, with light). Mirroring also swaps east and west, so walk the pergola again to undo it.
- **Inventory is weight-limited.** The route stashes things in the hub and drops them before each door.
- **Death is final.** Dying sends you to the ferryman at The River, and the game ends.
- **Seed-dependent details (seed 42):** the jeep radio is on channel 49, the diagram legend reads RD=POS BL=INF ST=GND WH=DET, and the radio chatter confirms that the "ground line" is the one to cut.
- **Trinity runs on a clock.** You arrive at 4:58:30 am and each turn is 15 seconds. Zero is 5:29:45. Wear the winged boots so each desert move takes one turn instead of five minutes.

The Walkthrough

1. Kensington Gardens, London (+15)

Nuclear war is minutes away. Take the gnomon from the sundial, win the umbrella and the paper crane, and ride the pram to the white door.

Command	What happens / why	Pts
VERBOSE, NE, UNSCREW GNOMON, TAKE GNOMON	The sundial's gnomon.	+5
SW, E, TAKE BALL, N, N, WAIT, THROW BALL AT UMBRELLA, TAKE UMBRELLA	The ball knocks the umbrella loose.	+6
W, W, S, TAKE COIN, GIVE COIN TO WOMAN, TAKE BAG, TAKE SMALL COIN, FEED BIRDS	Buy birdseed-style crumbs.	+1
E, WAIT, TAKE PAPER BIRD	The paper crane flutters down.	+3
NW, OPEN PERAMBULATOR, GET IN PERAMBULATOR, OPEN UMBRELLA, PUSH PERAMBULATOR S, GET IN PERAMBULATOR, OPEN UMBRELLA, TAKE ALL, S, E, E, (key)	Sail the pram across the Long Water and through the white door.	

2. The Hub and the Sundial (+7)

The splinter glows in the dark. The axe is stuck in the arbor. The gnomon has to be mirrored before it fits.

Command	What happens / why	Pts
N, NE, TAKE LOG, TAKE SPLINTER, SE, SE, W, N, U, TAKE AXE, S, D	Walking N/U/S/D through the pergola mirrors what you carry.	+2
E, NW, N, N, N, PUT GNOMON IN HOLE	A lever rises at the Vertex.	+5
S, S, S, SW, E, N, U, S, D, E, NW, N, N, N	Walk the pergola again to un-mirror yourself.	

3. Nagasaki (ring 5, Moor) (+7)

Give a schoolgirl the paper crane and umbrella. Her crane then carries you home. Back in the hub, fell the oak.

Command	What happens / why	Pts
TURN RING TO 5, WAIT ×7, PUSH LEVER, S, S, NE, SE, DROP BAG, DROP SMALL COIN, DROP AXE, ENTER DOOR	Stash items; the spade is heavy.	
OPEN UMBRELLA, TAKE ALL, E, TAKE SPADE, W, WAIT, GIVE PAPER TO GIRL, GIVE UMBRELLA TO GIRL, CLIMB BIRD		+4
ENTER DOOR, TAKE ALL, NW, W, CHOP OAK, PUSH OAK NORTH	The oak bridges the chasm to the Mesa.	+3

4. Nevada (ring 2, Ossuary) (+12)

Underground at the test site. Catch the skink, take the lantern and walkie-talkie, then take the icicle to the Crater.

Command	What happens / why	Pts
SW, E, N, N, PULL LEVER, TURN RING TO 2, WAIT ×7, PUSH LEVER, S, S, SW, NW, N, N, N	<i>The barrow wight lets you pass while you carry light.</i>	
SEARCH BONES, TAKE KEY, DROP BAG, DROP SPADE, DROP AXE, ENTER DOOR		+1
TAKE LANTERN, TURN ON LANTERN, W, DROP LANTERN, W	<i>The skink flees light into the crevice.</i>	+1
PUT SPLINTER IN CREVICE, TAKE SKINK, PUT SKINK IN POCKET	<i>Light in the crevice and the lantern behind leave it nowhere to hide.</i>	+3
E, TAKE WALKIE-TALKIE, TAKE LANTERN, E, E, TAKE AXE, TAKE SPADE, S, PUT KEY IN HOLE, TURN KEY, DOWN	<i>A spiked door blocks the barrow. The key opens a slope instead.</i>	+2
TURN OFF LANTERN, THROW SPADE AT ICICLES, TAKE ICICLE, E, E, E, N, N, S, S, NE, E, E	<i>Detour via the freezing Vertex so the icicle survives.</i>	+1
PUT ICICLE ON LUMP, DROP LANTERN, DROP WALKIE-TALKIE, TAKE LUMP	<i>The quenched lump is magnetic.</i>	+4

5. Earth Orbit (ring 1, Waterfall) (+3)

Travel in a soap bubble from the boy on the Promontory. The magpie's recipe needs a lizard "killed in the light of a crescent moon".

Command	What happens / why	Pts
W, W, SW, N, N, PULL LEVER, TURN RING TO 1, WAIT ×8, PUSH LEVER, S, S, NW, N, IN, WAIT ×3, S, SW, ENTER DOOR	<i>Ride the bubble through the door.</i>	
TAKE SKINK, SQUEEZE SKINK	<i>The crescent moon is only visible from orbit.</i>	+3
WAIT ×3, CUT FILM WITH AXE	<i>The lump anchors you to the satellite. Pop the bubble as the door comes round.</i>	

6. Eniwetok (ring 3, Mesa), Honey and Cage (+7)

Command	What happens / why	Pts
PUT SKINK IN POCKET, E, E, N, N, PULL LEVER, TURN RING TO 3, WAIT ×7, PUSH LEVER, S, S, NE, W, N, ENTER DOOR		
DOWN, OPEN BOX, PUSH BUTTON, S, NW, WAIT ×8, POINT AT COCONUT, TAKE COCONUT, SE, N, UP, ENTER DOOR	<i>The shot's blast knocks a coconut down.</i>	+3
S, E, REACH INTO HIVE, W, W, E, E, REACH INTO HIVE	<i>Second try, once the bees have calmed.</i>	+3
W, NE, OPEN DOOR, E, TAKE CAGE	<i>The cottage. The magpie in the cage recites the recipe.</i>	+1

7. The Cauldron (+7)

"Milk and honey, fresh whole lizard. Killed in the light of a crescent moon. Mix 'em with a pinch o' garlic. Then stand back!"

Command	What happens / why	Pts
OPEN BACK DOOR, E, SEARCH REFUSE, TAKE GARLIC, W		+1
TAKE SKINK, PUT SKINK IN CAULDRON, PUT GARLIC IN CAULDRON		+2
DROP COCONUT, HIT COCONUT WITH AXE, TAKE COCONUT, POUR MILK INTO CAULDRON, PUT HAND IN CAULDRON	<i>The honey washes off your hand into the pot.</i>	+3
DROP COCONUT, W, E, TAKE EMERALD	<i>Boom! An emerald waits in the cauldron.</i>	+1

8. Siberia (ring 4, Herb Garden) (+1)

Command	What happens / why	Pts
W, SW, SW, E, N, N, PULL LEVER, TURN RING TO 4, WAIT ×11, PUSH LEVER, S, S, NE, W, NE, E, E, ENTER WHITE DOOR		
DOWN, NE, NE, OPEN CAGE, EXAMINE FISSURE, TAKE LEMMING, PUT LEMMING IN CAGE, CLOSE CAGE	<i>Free the magpie and cage the lemming trapped in the fissure.</i>	+1
SW, SW, UP, ENTER DOOR	<i>Leave before the countdown ends.</i>	

9. Wabewalker's Crypt (+6)

Command	What happens / why	Pts
W, W, SW, SW, W, W, TAKE SPADE, E, N, OPEN CRYPT WITH SPADE	Prise off the lid.	+3
DROP SPADE, REMOVE SHROUD, PUT EMERALD IN GREEN BOOT, REMOVE BANDAGE, TAKE COIN FROM MOUTH	Winged boot, plus the ferryman's silver coin.	+3
WEAR SHROUD, DROP BANDAGE, DROP AXE, TAKE GREEN BOOT, TAKE RED BOOT, WEAR GREEN BOOT, WEAR RED BOOT		

10. Crumbs, the Ferry and the Islet (ring 6) (+0)

The roadrunner loves crumbs. Recover the bag from the Ossuary, then cross The River disguised as a shade.

Command	What happens / why	Pts
S, SE, NE, NE, E, E, TAKE LANTERN, TAKE WALKIE-TALKIE, W, W, SW, SW, NW, N, TURN ON LANTERN, N, N, TAKE BAG, S, TURN KEY, DOWN, TURN OFF LANTERN, DROP SMALL COIN, E		
E, E, N, N, PULL LEVER, TURN RING TO 6, WAIT ×8, PUSH LEVER, S, S, SE, WAIT, WAIT		
GET IN DORY, GIVE SILVER COIN TO OARSMAN, S, ENTER DOOR	The shroud convinces the oarsman. The roadrunner leads you through the door.	

11. Trinity: The Tower and the Ranch (+11)

4:58:30 am, in the shack atop the 100-foot tower. Stay inside until the jeep leaves at 5:00.

Command	What happens / why	Pts
DROP LANTERN, TAKE BOOK, READ BOOK, TAKE CARDBOARD, DROP BOOK, TAKE LANTERN, WAIT, W, D, D		+1
EXAMINE DIAGRAM, DROP CARDBOARD, TAKE RUBY, PUT RUBY IN RED BOOT, DROP WALKIE-TALKIE	The diagram gives the wire legend. The roadrunner drops the ruby.	+5
SE, SE, E, S, U, OPEN LEFT DOOR, W, N, N, CLOSE DOOR, OPEN CAGE, OPEN DOOR	Hide in the closet. The lemming runs straight into the rattlesnake.	+3
PUT BAG IN CAGE, CLOSE CAGE, S, LOOK UNDER PAPER, TAKE SCREWDRIVER, S, W, TAKE KNIFE	The closed cage keeps the roadrunner out of the crumbs.	+2

12. Trinity: Binoculars, Key, Radio (+16)

Command	What happens / why	Pts
E, E, E, SE, NE, DROP CAGE, DROP SCREWDRIVER, DROP KNIFE, TURN ON LANTERN, U, TAKE BINOCULARS	The rotten landing collapses into the reservoir.	
D, TAKE ALL, U, OUT, DROP LANTERN, D, NE, TAKE ALL	The waterproof lantern finds them underwater.	+1
SW, W, SW, W, S, S, FOCUS BINOCULARS ON ENTRANCE, WAIT, ROADRUNNER, GET KEY, TAKE KEY	Spot the key in the shelter. The "gray torpedo" fetches it.	+7
DROP BINOCULARS, N ×4, UNLOCK PADLOCK WITH KEY, TAKE PADLOCK, DROP PADLOCK, OPEN BOX, TAKE WALKIE-TALKIE		+1
EXAMINE WATCH, NW, NW, NW, IN, EXAMINE DIAL, TURN ON WALKIE-TALKIE, PULL ANTENNA, SET SLIDER TO 49, OUT	Listen in on the test crew's channel.	+6
SE, SE, SE, OPEN BREAKER, CLOSE BREAKER	"Lotta crossed fingers up here."	+1

13. Trinity: Sabotage (+8)

Observers watch the tower. After the first descent you can only afford one more unlit turn on it. Distract the searchlight with the dog.

Command	What happens / why	Pts
SW ×4, OPEN CAGE, TAKE BAG, DROP BAG	The roadrunner pecks the crumbs for a few turns, then wakes the German shepherd.	
N, NE, E, U, U, E	The searchlight swings toward the barking as you climb.	
UNSCREW PANEL WITH SCREWDRIVER, PULL CHAIN, WAIT ×12	Wait for the auto-sequencer so the crew won't scrub.	+3
CUT STRIPED WIRE WITH KNIFE	ST=GND. The multigigaton gadget fizzles to history's 20 kilotons, and you wake at Palace Gate.	+5

Where the Points Come From

Action	Pts	Action	Pts
take gnomon	5	hit coconut with axe	1
take ball	1	pour milk into cauldron	1
take umbrella	5	put hand in cauldron	1
give coin to woman	1	take emerald	1
take paper bird	3	put lemming in cage	1
take splinter	1	open crypt with spade	3
take axe	1	remove shroud	1
put gnomon in hole	5	put emerald in green boot	1
take spade	1	take coin from mouth	1
give umbrella to girl	3	take cardboard	1
push oak north	3	examine diagram	3
take (skeleton) key	1	take ruby	1
take lantern	1	put ruby in red boot	1
take skink	3	lemming meets the rattlesnake	3
take walkie-talkie	1	take screwdriver	1
turn key	1	take knife	1
take icicle	1	recover binoculars	1
put icicle on lump	3	focus binoculars on entrance	3
take lump	1	roadrunner, get key	3
squeeze skink (orbit)	3	take (steel) key	1
take coconut	3	unlock padlock	1
reach into hive	3	examine dial	3
take cage	1	set slider to 49	3
take garlic	1	close breaker	1
put skink in cauldron	1	unscrew panel	3
put garlic in cauldron	1	cut striped wire	5

Opcode Census

Every instruction executed during the 600-move run was decoded with the interpreter's own disassembler and counted by mnemonic. The run used **61 distinct opcodes** over **986,642** instructions. Trinity is the only walkthrough story so far to use the V4-only `read_char`, `scan_table`, `call_vs` with more than three arguments, and the window/cursor opcodes for its upper-window status bar and epigraph boxes.

#	Opcode	Form:No.	Executions	Purpose
1	jz	1OP:128	141,001	branch if zero
2	loadb	2OP:16	96,866	load byte from array
3	storeb	VAR:226	84,615	store byte into array
4	je	2OP:1	76,489	branch if equal to any operand
5	add	2OP:20	67,007	signed addition
6	loadw	2OP:15	63,181	load word from array
7	dec_chk	2OP:4	55,533	decrement and branch if less
8	store	2OP:13	52,415	store to variable
9	storew	VAR:225	35,021	store word into array
10	mul	2OP:22	31,115	signed multiplication
11	jump	1OP:140	26,331	unconditional jump
12	jg	2OP:3	25,915	branch if greater (signed)
13	inc_chk	2OP:5	24,375	increment and branch if greater
14	test_attr	2OP:10	23,845	branch if object has attribute
15	call_2s	2OP:25	16,477	call routine with 1 arg, store
16	call_vs	VAR:224	15,755	call routine with up to 3 args, store
17	scan_table	VAR:247	12,829	search a table (V4+)
18	inc	1OP:133	12,503	increment variable
19	jle	2OP:2	11,286	branch if less (signed)
20	get_sibling	1OP:129	9,252	next sibling, branch if any
21	get_prop_add r	2OP:18	9,249	address of property data
22	get_child	1OP:130	8,776	first child object, branch if any
23	rtrue	0OP:176	7,996	return true
24	call_1s	1OP:136	7,961	call routine, no args, store result
25	sub	2OP:21	7,230	signed subtraction
26	get_parent	1OP:131	5,446	parent object
27	test	2OP:7	4,861	branch if all bitmap flags set

#	Opcode	Form:No.	Executions	Purpose
28	div	2OP:23	4,771	signed division
29	get_prop_len	1OP:132	4,501	length of property data
30	ret	1OP:139	4,453	return value
31	print	0OP:178	4,167	print inline string
32	jin	2OP:6	3,852	branch if object is a child of
33	print_char	VAR:229	3,765	print ZSCII character
34	get_prop	2OP:17	3,135	read property value
35	rfalse	0OP:177	2,815	return false
36	print_obj	1OP:138	2,489	print object short name
37	and	2OP:9	2,237	bitwise AND
38	new_line	0OP:187	2,219	print newline
39	push	VAR:232	1,822	push onto stack
40	set_text_style	VAR:241	1,801	bold / reverse / italic / roman
41	clear_attr	2OP:12	1,767	clear object attribute
42	insert_obj	2OP:14	1,736	move object into another
43	print_paddr	1OP:141	1,163	print string at packed address
44	ret_popped	0OP:184	1,124	return top of stack
45	set_attr	2OP:11	1,024	set object attribute
46	dec	1OP:134	814	decrement variable
47	output_stream	VAR:243	572	select/deselect output streams (memory stream)
48	set_cursor	VAR:239	515	position cursor in upper window
49	set_window	VAR:235	506	select upper or lower window
50	buffer_mode	VAR:242	505	toggle word-wrap buffering
51	read	VAR:228	493	read a line of input (sread)
52	random	VAR:231	456	random number / reseed
53	print_ret	0OP:179	294	print inline string, newline, return true
54	print_num	VAR:230	111	print signed number
55	remove_obj	1OP:137	82	detach object from parent
56	put_prop	VAR:227	53	write property value
57	sound_effect	VAR:245	52	bleeps and sound control
58	split_window	VAR:234	8	size the upper window

#	Opcode	Form:No.	Executions	Purpose
59	erase_window	VAR:237	4	clear a window
60	print_addr	1OP:135	4	print string at byte address
61	read_char	VAR:246	2	read a single keypress (V4+)

Full Command List

One line per input, exactly as replayed by the test (seed 42). "key" answers a press-any-key screen (read_char). Read down each column.

1 key	125 drop spade	249 take cage	373 get in dory
2 verbose	126 drop axe	250 open back door	374 give silver coin to oarsm
3 northeast	127 enter door	251 east	375 south
4 unscrew gnomon	128 take lantern	252 search refuse	376 enter door
5 take gnomon	129 turn on lantern	253 take garlic	377 drop lantern
6 southwest	130 west	254 west	378 take book
7 east	131 drop lantern	255 take skink	379 read book
8 take ball	132 west	256 put skink in cauldron	380 take cardboard
9 north	133 put splinter in crevice	257 put garlic in cauldron	381 drop book
10 north	134 take skink	258 drop coconut	382 take lantern
11 wait	135 put skink in pocket	259 hit coconut with axe	383 wait
12 throw ball at umbrella	136 east	260 take coconut	384 west
13 take umbrella	137 take walkie-talkie	261 pour milk into cauldron	385 down
14 west	138 take lantern	262 put hand in cauldron	386 down
15 west	139 east	263 drop coconut	387 examine diagram
16 south	140 east	264 west	388 drop cardboard
17 take coin	141 take axe	265 east	389 take ruby
18 give coin to woman	142 take spade	266 take emerald	390 put ruby in red boot
19 take bag	143 south	267 west	391 drop walkie-talkie
20 take small coin	144 put key in hole	268 southwest	392 southeast
21 feed birds	145 turn key	269 southwest	393 southeast
22 east	146 down	270 east	394 east
23 wait	147 turn off lantern	271 north	395 south
24 take paper bird	148 throw spade at icicles	272 north	396 up
25 northwest	149 take icicle	273 pull lever	397 open left door
26 open perambulator	150 east	274 turn ring to 4	398 west
27 get in perambulator	151 east	275 wait	399 north
28 open umbrella	152 east	276 wait	400 north
29 push perambulator south	153 north	277 wait	401 close door
30 get in perambulator	154 north	278 wait	402 open cage
31 open umbrella	155 south	279 wait	403 open door
32 take all	156 south	280 wait	404 put bag in cage
33 south	157 northeast	281 wait	405 close cage
34 east	158 east	282 wait	406 south
35 east	159 east	283 wait	407 look under paper
36 key	160 put icicle on lump	284 wait	408 take screwdriver
37 north	161 drop lantern	285 wait	409 south
38 northeast	162 drop walkie-talkie	286 push lever	410 west
39 take log	163 take lump	287 south	411 take knife
40 take splinter	164 west	288 south	412 east
41 southeast	165 west	289 northeast	413 east
42 southeast	166 southwest	290 west	414 east
43 west	167 north	291 northeast	415 southeast
44 north	168 north	292 east	416 northeast
45 up	169 pull lever	293 east	417 drop cage
46 take axe	170 turn ring to 1	294 enter white door	418 drop screwdriver
47 south	171 wait	295 down	419 drop knife
48 down	172 wait	296 northeast	420 turn on lantern
49 east	173 wait	297 northeast	421 up
50 northwest	174 wait	298 open cage	422 take binoculars
51 north	175 wait	299 examine fissure	423 down
52 north	176 wait	300 take lemming	424 take all
53 north	177 wait	301 put lemming in cage	425 up
54 put gnomon in hole	178 wait	302 close cage	426 out
55 south	179 push lever	303 southwest	427 drop lantern
56 south	180 south	304 southwest	428 down
57 south	181 south	305 up	429 northeast
58 southwest	182 northwest	306 enter door	430 take all
59 east	183 north	307 west	431 southwest
60 north	184 in	308 west	432 west
61 up	185 wait	309 southwest	433 southwest
62 south	186 wait	310 southwest	434 west
63 down	187 wait	311 west	435 south
64 east	188 south	312 west	436 south
65 northwest	189 southwest	313 take spade	437 focus binoculars on entra
66 north	190 enter door	314 east	438 wait
67 north	191 take skink	315 north	439 roadrunner, get key
68 north	192 squeeze skink	316 open crypt with spade	440 take key
69 turn ring to 5	193 wait	317 drop spade	441 drop binoculars
70 wait	194 wait	318 remove shroud	442 north
71 wait	195 wait	319 put emerald in green boot	443 north
72 wait	196 cut film with axe	320 remove bandage	444 north
73 wait	197 put skink in pocket	321 take coin from mouth	445 north
74 wait	198 east	322 wear shroud	446 unlock padlock with key
75 wait	199 east	323 drop bandage	447 take padlock
76 wait	200 north	324 drop axe	448 drop padlock
77 push lever	201 north	325 take green boot	449 open box
78 south	202 pull lever	326 take red boot	450 take walkie-talkie
79 south	203 turn ring to 3	327 wear green boot	451 examine watch
80 northeast	204 wait	328 wear red boot	452 northwest
81 southeast	205 wait	329 south	453 northwest
82 drop bag	206 wait	330 southeast	454 northwest
83 drop small coin	207 wait	331 northeast	455 in
84 drop axe	208 wait	332 northeast	456 examine dial

85	enter door	209	wait	333	east	457	turn on walkie-talkie
86	open umbrella	210	wait	334	east	458	pull antenna
87	take all	211	push lever	335	take lantern	459	set slider to 49
88	east	212	south	336	take walkie-talkie	460	out
89	take spade	213	south	337	west	461	southeast
90	west	214	northeast	338	west	462	southeast
91	wait	215	west	339	southwest	463	southeast
92	give paper to girl	216	north	340	southwest	464	open breaker
93	give umbrella to girl	217	enter door	341	northwest	465	close breaker
94	climb bird	218	down	342	north	466	southwest
95	enter door	219	open box	343	turn on lantern	467	southwest
96	take all	220	push button	344	north	468	southwest
97	northwest	221	south	345	north	469	southwest
98	west	222	northwest	346	take bag	470	open cage
99	chop oak	223	wait	347	south	471	take bag
100	push oak north	224	wait	348	turn key	472	drop bag
101	southwest	225	wait	349	down	473	north
102	east	226	wait	350	turn off lantern	474	northeast
103	north	227	wait	351	drop small coin	475	east
104	north	228	wait	352	east	476	up
105	pull lever	229	wait	353	east	477	up
106	turn ring to 2	230	wait	354	east	478	east
107	wait	231	point at coconut	355	north	479	unscrew panel with screwd
108	wait	232	take coconut	356	north	480	pull chain
109	wait	233	southeast	357	pull lever	481	wait
110	wait	234	north	358	turn ring to 6	482	wait
111	wait	235	up	359	wait	483	wait
112	wait	236	enter door	360	wait	484	wait
113	wait	237	south	361	wait	485	wait
114	push lever	238	east	362	wait	486	wait
115	south	239	reach into hive	363	wait	487	wait
116	south	240	west	364	wait	488	wait
117	southwest	241	west	365	wait	489	wait
118	northwest	242	east	366	wait	490	wait
119	north	243	east	367	push lever	491	wait
120	north	244	reach into hive	368	south	492	wait
121	north	245	west	369	south	493	cut striped wire with kni
122	search bones	246	northeast	370	southeast	494	wait
123	take key	247	open door	371	wait	495	score
124	drop bag	248	east	372	wait		